🪣 1. Bucket as a Data Structure Concept

A **bucket** is simply a **container (like a list or a group)** used to hold elements that share a common property.

For example, in a **hash table**, a *bucket* stores items that hash to the same index.

Example – Hashing with Buckets:

```
# Suppose we want to group numbers by their remainder when divided by 3


numbers = [1, 4, 6, 8, 9, 12, 15]

buckets = {0: [], 1: [], 2: []}


for num in numbers:
    remainder = num % 3
    buckets[remainder].append(num)


print(buckets)
```

Output:

{0: [6, 9, 12, 15], 1: [1, 4, 7, 10], 2: [8]}

Here each key (0, 1, 2) represents a **bucket**, and the values are lists of numbers.

🧱 2. Bucket in Bucket Sort

In the **bucket sort algorithm**, data is divided into **multiple buckets**, sorted individually, and then combined.

Example:

```
def bucket_sort(nums):
    # Create empty buckets
    buckets = [[] for _ in range(10)]
```

```
    # Distribute elements into buckets
    for num in nums:
        index = int(num * 10)
        buckets[index].append(num)

    # Sort each bucket and combine
    sorted_list = []
    for bucket in buckets:
        sorted_list.extend(sorted(bucket))

    return sorted_list

data = [0.42, 0.32, 0.23, 0.52, 0.25, 0.47, 0.51]
print(bucket_sort(data))
```

Output:

```
[0.23, 0.25, 0.32, 0.42, 0.47, 0.51, 0.52]
```

Each list inside buckets is a **bucket** of elements within a range.

☁️ 3. Bucket in Cloud Storage (e.g., AWS S3, GCP, Azure)

When using Python to work with **cloud services**, a *bucket* means a **storage container** for files.

For example, in **AWS S3**, a bucket stores objects (files):

```
import boto3

s3 = boto3.resource('s3')
bucket = s3.Bucket('my-data-bucket')
```

```
for file in bucket.objects.all():
    print(file.key)
```

Here, "my-data-bucket" is a **cloud bucket**.

✅ **Summary**

Context	Meaning of “Bucket”	Example
Data Structures	A group or container for similar items	Buckets in hash table
Sorting Algorithm	Used to divide data for sorting	Bucket Sort
Cloud Storage	A file container (e.g., AWS S3 bucket)	s3.Bucket('my-bucket')